
Reliable and Trustworthy Artificial Intelligence

Assignment #4

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Instructions

- **Release date:** Tuesday, June 2, 2026
- **Due date:** Friday, June 12, 2026, by 11:59 PM
- **Submission:** Upload the project to GitHub and submit the link to Uclass. Late submissions will not be accepted, either through the system or email.

Important:

- Reproducibility is crucial for grading. Your GitHub project should include a `requirements.txt` (or a Dockerfile) that describes all external modules and environment setup.
- Please include comments in your code to help me understand it.
- Include a `test.py` file that demonstrates running α, β -Crown on your chosen model. The specific format is not required, but it should demonstrate your code's validity.

1 Overview

In this assignment, you will familiarize yourself with α, β -Crown, a state-of-the-art neural network verifier based on bound propagation techniques. While Marabou (Assignment #3) uses an SMT-based approach, α, β -Crown uses **linear relaxation** and **branch-and-bound** to compute tight output bounds, often achieving significantly faster verification on larger networks. You will set up the tool, explore its built-in models, and run it on an external model and dataset of your choosing.

Reference: α, β -Crown is the winner of multiple VNN-COMP (Verification of Neural Networks Competition) iterations. The key papers are “Fast and Complete: Enabling Complete Neural Network Verification with Rapid and Massively Parallel Incomplete Verifiers” (Wang et al., ICLR 2021) and subsequent works. Reading the tool’s documentation will help you understand its configuration system.

2 Problem 1: Explore the α, β -Crown Models Directory

Background

α, β -Crown provides a collection of pre-defined models and verification configurations in its repository. Understanding what is available will help you choose an appropriate external model for the next part.

What to do

1. Visit the α, β -Crown GitHub repository:
<https://github.com/Verified-Intelligence/alpha-beta-CROWN>
2. Navigate to the models directory:
https://github.com/Verified-Intelligence/alpha-beta-CROWN/tree/main/complete_verifier/models
3. Review the files available and document:
 - What types of models are provided (architecture, format)?
 - What verification configurations (YAML files) are available?
 - How does α, β -Crown’s model specification differ from Marabou’s?

Tip: α, β -Crown uses YAML configuration files to specify models, datasets, verification properties, and solver parameters. Understanding this configuration system is key to running the tool successfully. Look at the example configs in the `complete_verifier/exp_configs/` directory.

3 Problem 2: Run α, β -Crown with an External Model and Dataset

Background

To demonstrate your understanding of α, β -Crown, you will apply it to a model that is *not* already included in the models directory. This requires understanding the tool’s configuration format, preparing a compatible model, and interpreting the verification results.

What to do

Choose an external model and dataset, configure α, β -Crown, and run a verification query.

Step-by-step guide

1. Install α, β -Crown.

- Clone the repository and follow the installation instructions (a Conda environment is recommended).
- Verify the installation by running one of the provided example configurations.
- Document any issues encountered during installation and how you resolved them.

2. Select an external model and dataset.

- Choose a model that is *not* already in the α, β -Crown models directory.
- The model should be in ONNX or PyTorch format.
- Start with a small model — α, β -Crown can handle larger networks than Marabou, but very large models will still be slow for complete verification.
- If you used Marabou in Assignment #3, consider using the *same* model to compare the two tools' performance.

3. Create a configuration file.

- Write a YAML configuration file specifying:
 - The model path and architecture.
 - The dataset and input specification.
 - The verification property (e.g., robustness within an ℓ_∞ -ball of radius ε).
 - Solver parameters (e.g., timeout, branching strategy).
- Use the provided example configs as templates.

4. Run α, β -Crown and interpret results.

- Run the verifier using your configuration file.
- Record for each verification instance whether the property was:
 - **Verified** (the property holds for all inputs in the region).
 - **Falsified** (a counterexample was found).
 - **Timeout** (verification did not complete within the time limit).
- Report verification times and compare with Marabou if applicable.

Tip: If you used Marabou in Assignment #3, comparing the two tools on the same model and property provides valuable insight. Consider discussing differences in verification time, supported model formats, and ease of configuration in your report.

4 Deliverables & Grading

4.1 Exploration report (20%)

A summary of the α, β -Crown models directory: what models, configurations, and verification setups are available. If applicable, briefly compare the organization with Marabou's resources.

4.2 Implementation & results (50%)

1. Your code and configuration files for running α, β -Crown on the external model.
2. A clear record of:
 - Which model and dataset you chose, and why.
 - Any modifications or preprocessing steps required for compatibility.
 - The verification results (verified/falsified/timeout), runtimes, and any counterexamples found.

4.3 Report (30%)

Submit a short PDF report (**1–2 pages**) that includes:

1. A description of your model, dataset, and verification property.
2. The results and their interpretation.
3. A comparison between α, β -Crown and Marabou (if you used the same model): differences in speed, scalability, ease of use, and supported features.
4. A brief discussion of α, β -Crown's strengths and limitations based on your experience.

On the use of AI tools

You may use AI assistants (e.g., ChatGPT, GitHub Copilot) as a *learning aid*, but you must demonstrate genuine understanding of your work. To ensure this:

- Your GitHub repository must show a **meaningful commit history** that reflects incremental development — not a single bulk commit. We will inspect the git log.
- The report must include your **own interpretation and reasoning**, not generic descriptions. Discuss specific behaviors you observed in your experiments, including any unexpected results.

5 Submission Guidelines

Submission Checklist

Upload the following to GitHub and submit the repository link to Uclass:

1. `requirements.txt` (or `Dockerfile` / `environment.yml`) — environment setup and dependencies.
2. `test.py` — runs α, β -Crown on your model and produces verification results.
3. YAML configuration file(s) used for verification.
4. `report.pdf` — your analysis report (1–2 pages).
5. `README.md` — brief instructions on how to install α, β -Crown and run your code.